Triangle meshes

Face array

- OK for “triangle soups”. Exemple: STL
- Vertex duplication (**very** verbose)
- No explicit connectivity information (traversals are **hard**)

Triangles		
x_{11} y_{11} z_{11}	x_{12} y_{12} z_{12}	x_{13} y_{13} z_{13}
x_{21} y_{21} z_{21}	x_{22} y_{22} z_{22}	x_{23} y_{23} z_{23}
...	...	...
x_{F1} y_{F1} z_{F1}	x_{F2} y_{F2} z_{F2}	x_{F3} y_{F3} z_{F3}

Face array + vertex array

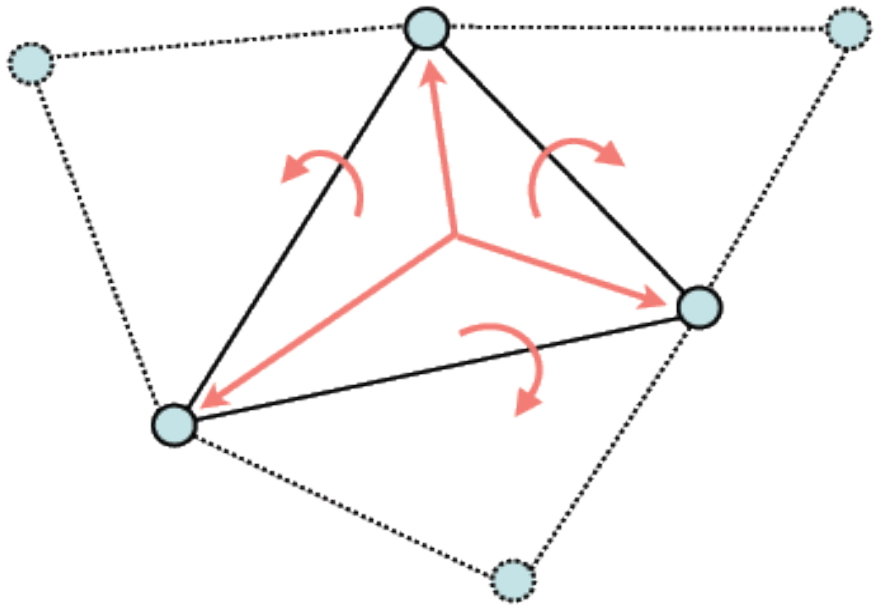
- Example: format OBJ, OFF, PLY
- No vertex duplication
- No neighbor info

Vertices
x_1 y_1 z_1
...
x_V y_V z_V

Triangles
v_{11} v_{12} v_{13}
...
...
...
...
v_{F1} v_{F2} v_{F3}

Face-based connectivity

- Vertex:
  - Position (x,y,z)
  - 1 Face(ref)

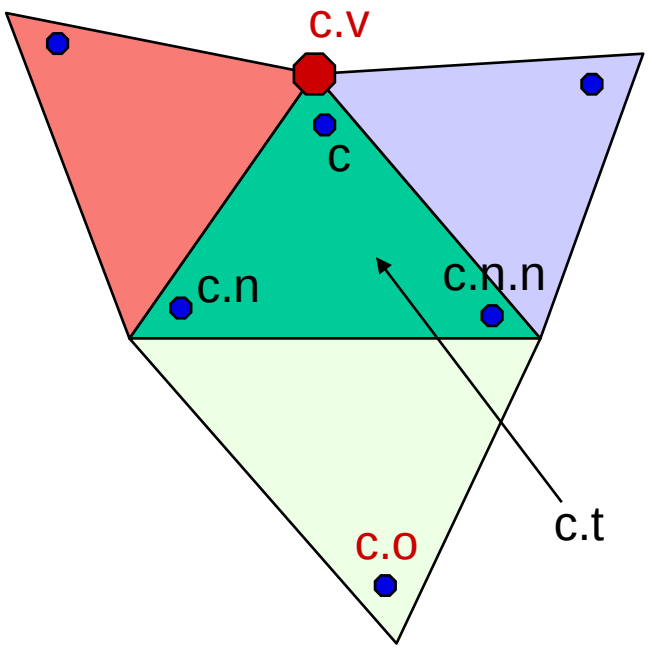


- **Face**:
  - 3 vertices (refs)
  - 3 faces (refs)

- Edges are not explicitly represented

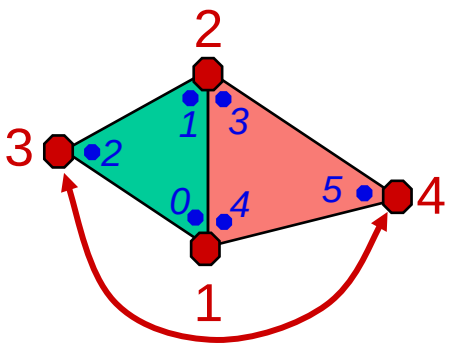
Corner-based connectivity

- For each corner:
  - Index to position table
  - Index of opposite corner
- Make corners consecutive

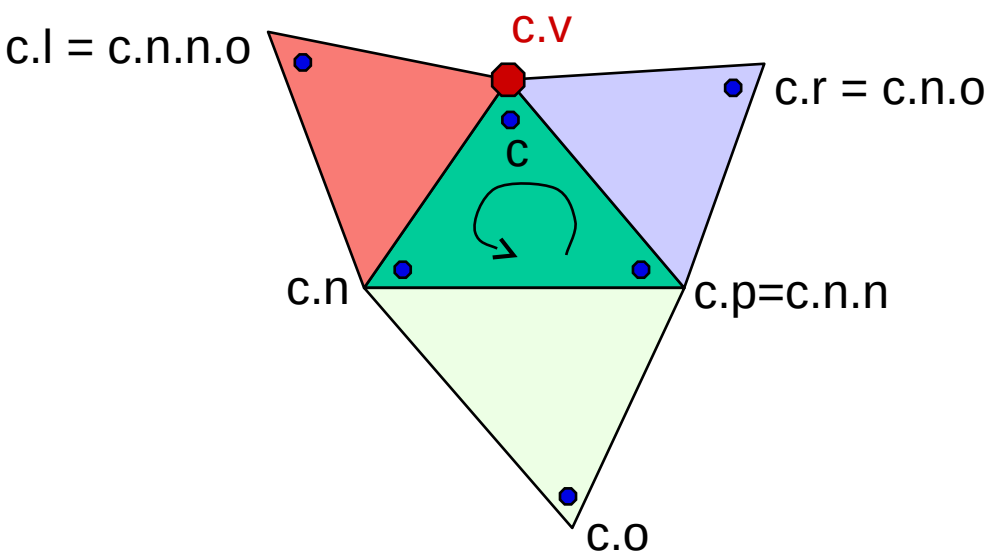


	v	o	a
Triangle 0 corner 0	1	7	a
Triangle 0 corner 1	2	8	b
Triangle 0 corner 2	3	5	c
Triangle 1 corner 3	2	9	c
Triangle 1 corner 4	1	6	d
Triangle 1 corner 5	4	2	e

vertex 1	x	y	z
vertex 2	x	y	z
vertex 3	x	y	z
vertex 4	x	y	z

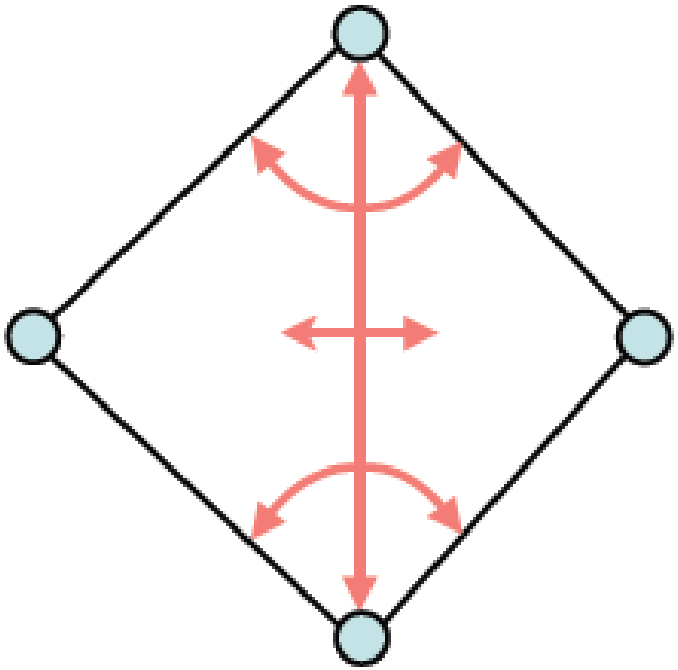


Can navigate easily:



Edge-based connectivity

- Vertex:
  - position
  - 1 edge
- **Edge**:
  - 2 vertices
  - 2 faces
  - 4 edges
- Face:
  - 1 edge

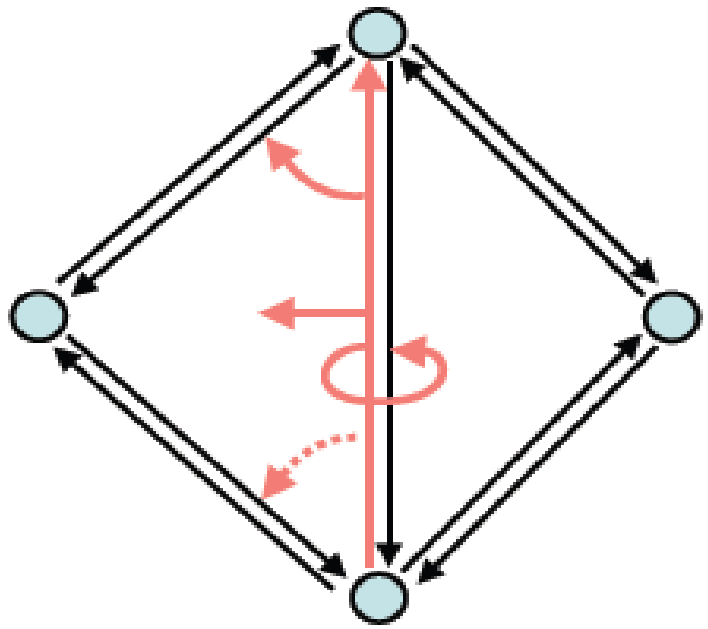


Half-edge connectivity

- Vertex:
  - position
  - 1 half-edge (outgoing)

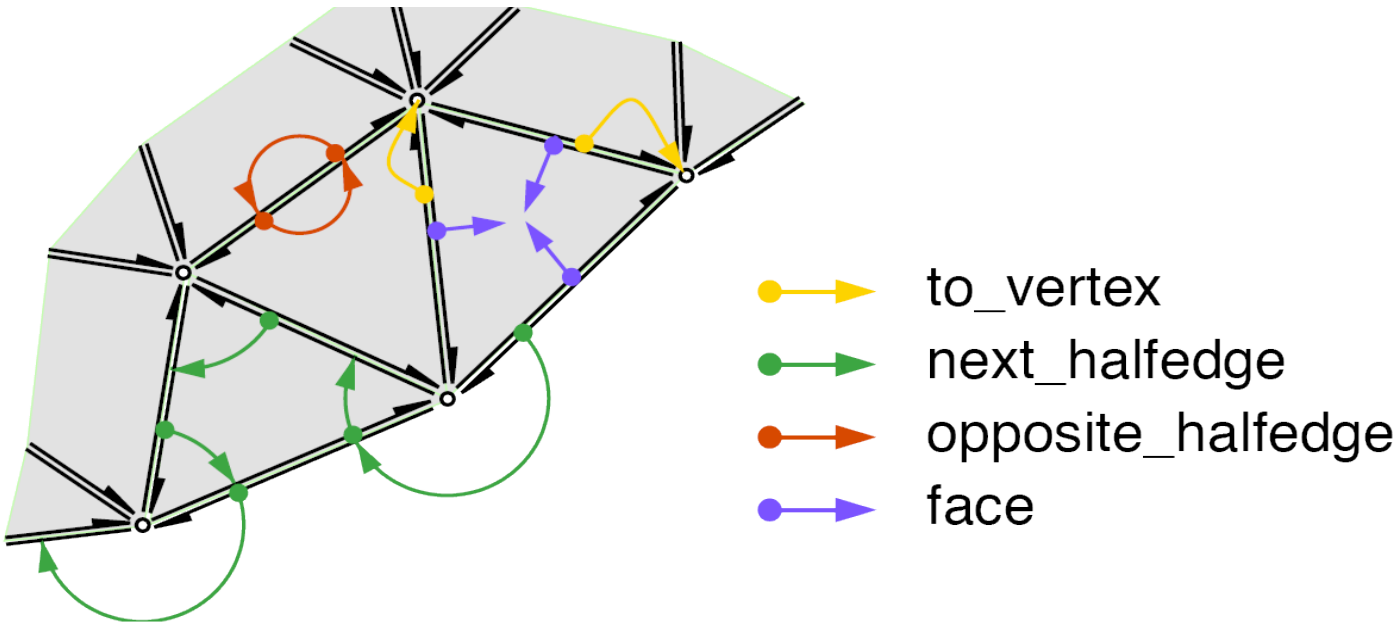
- **Half-edge**:
  - 1 vertex (endpoint)
  - 1 face
  - 2 or 3 half-edges
    next, opposite, [previous]

- Face:
  - 1 half-edge

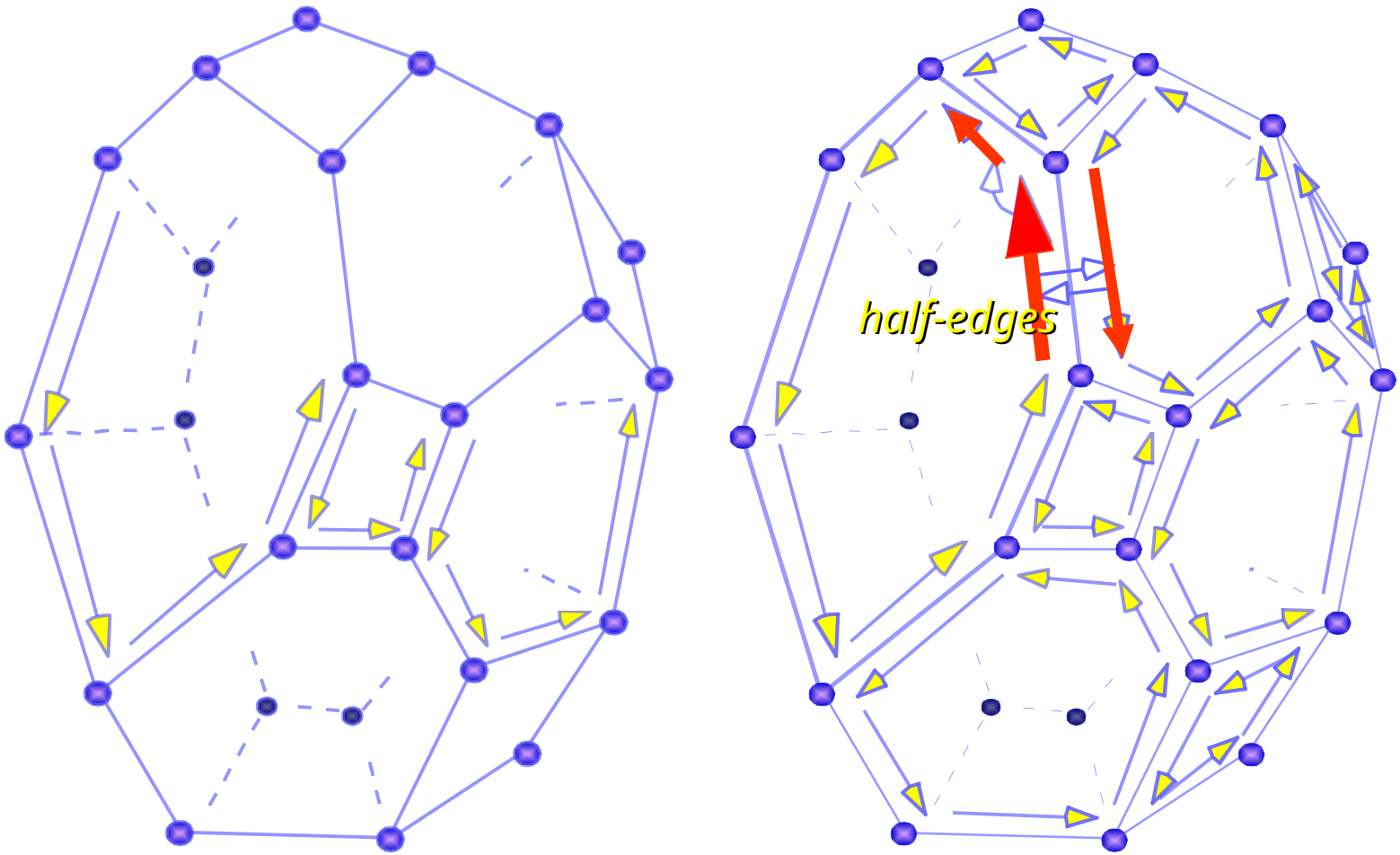


Half-edge connectivity

- **Half-edge**:
  - *to_vertex:* vertex pointed to by the half-edge
  - *next_halfedge: next half-edge* of the face (or the boundary)
  - *opposite_halfedge*
  - *Face:* face the half-edge belongs to (null if boundary)



Not limited to triangle meshes



Half-edge connectivity

Halfedge data structure

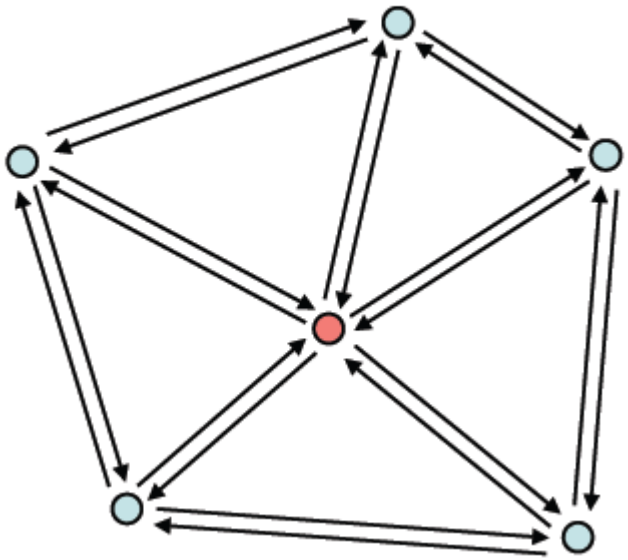
```
struct Halfedge {
  HalfedgeRef      next_halfedge;
  HalfedgeRef      opposite_halfedge;
  FaceRef          face;
  VertexRef        to_vertex;
};
```

```
struct Face {
  HalfedgeRef      halfedge;
};

struct Vertex {
  HalfedgeRef      outgoing_halfedge;
};
```

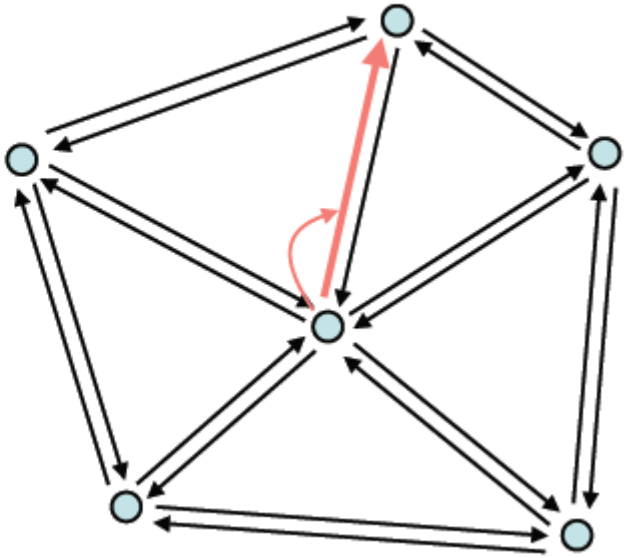
Traversing the 1-ring

1. Start at vertex



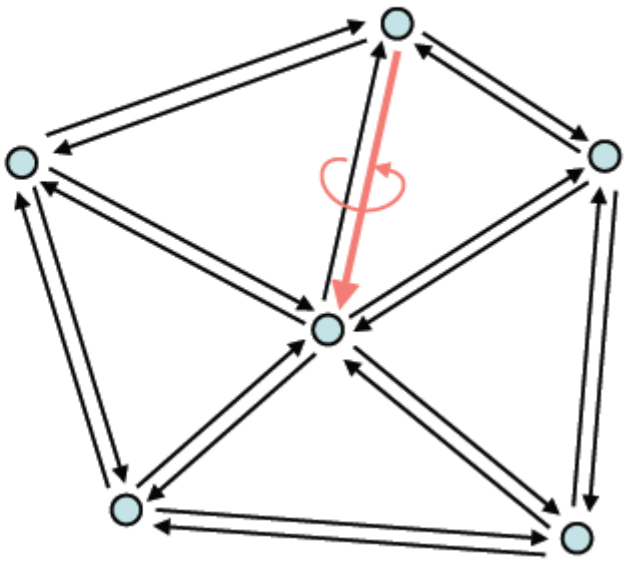
Traversing the 1-ring (CW)

1. Start at vertex
2. Outgoing halfedge



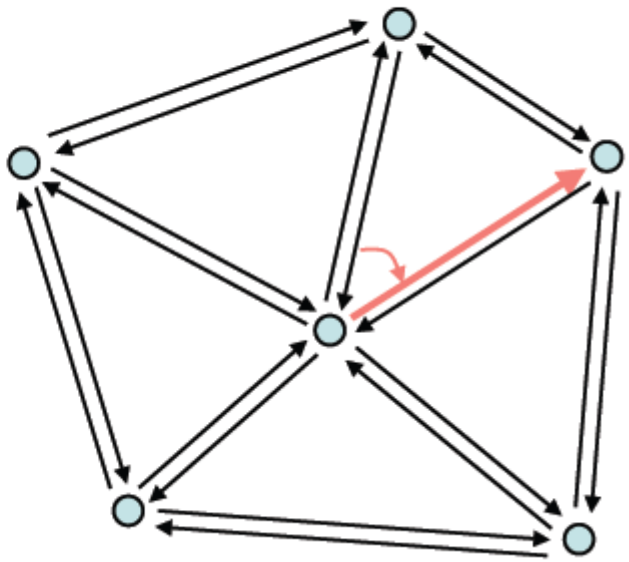
Traversing the 1-ring

1. Start at vertex
2. Outgoing halfedge
3. Opposite halfedge



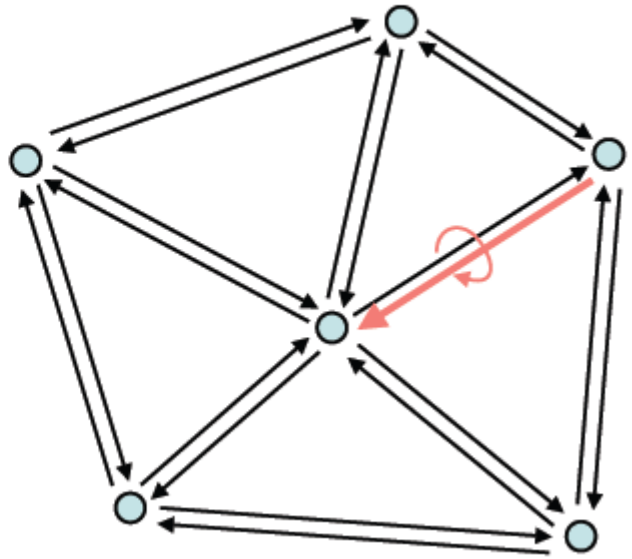
Traversing the 1-ring

1. Start at vertex
2. Outgoing halfedge
3. Opposite halfedge
4. Next halfedge



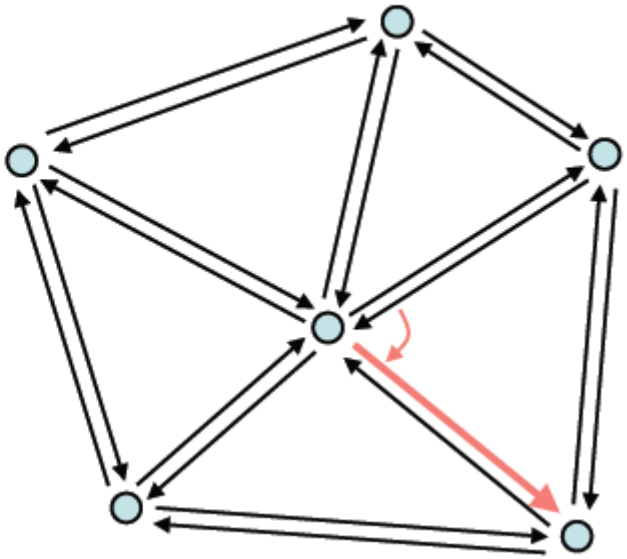
Traversing the 1-ring

1. Start at vertex
2. Outgoing halfedge
3. Opposite halfedge
4. Next halfedge
5. Opposite

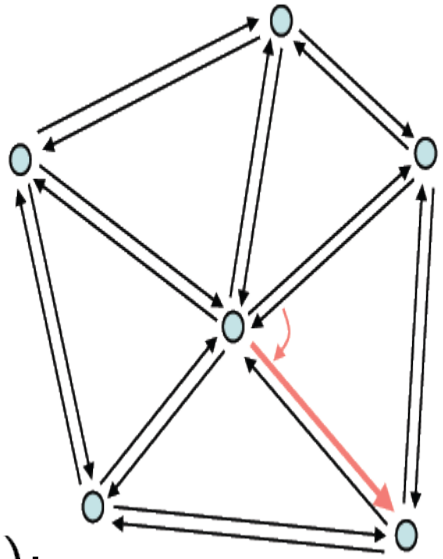


Traversing the 1-ring

1. Start at vertex
2. Outgoing halfedge
3. Opposite halfedge
4. Next halfedge
5. Opposite
6. Next
7. ...



Traversing the 1-ring



```
enumerate_1_ring(Vertex * center)
{
    HalfedgeRef h = outgoing_halfedge(center);
    HalfedgeRef hstop = h;
    do {
        VertexRef v = to_vertex(h);
        // do something with v
        h = next_halfedge(opposite_halfedge(h));
    } while ( h != hstop );
}
```

Half-edge connectivity

Implementation

- One **face array**, one **half-edge array**, and one **vertex array**
- References = indices (better than pointers)
  - Easier memory handling
  - A single index for all the attibutes of an element
- Each pair of half-edges are stored in consecutive positions in the array
  - Opposite_halfedge is implicit (saving space)
    - h **mod** 2 = 0 → opp:=h+1
      h **mod** 2 = 1 → opp:= h-1
  - Attributes can be attached to faces, edges, half-edges and vertices

Half-edge connectivity

Summary of *half-edge data structure*:

- Suitable for any two-manifold (with/without boundary)
- Not limited to triangular faces (OK for arbitrary polygons)
- Easy to associate dynamic attributes to the different elements: per-face, per-vertex, per-edge and per-corner attributes

Mesh APIs

Important topics

- **Access**: how are elements accessed? How are neighboring elements enumerated?
- **Modification**: how can we add/delete elements? How can we guarantee mesh consistency after the updates?
- **Dynamic attribs**: How can we associate attributes to the different elements?
- **Input/output:** how can we generate the model from a vertex array + face array?

Mesh APIs

- CGAL
  - `www.cgal.org`
  - Computational geometry
  - Free for non-commercial use

- OpenMesh
  - `www.openmesh.org`
  - Mesh processing
  - Free, LGPL licence

And many others… (VCGlib, Triangle, trimesh,...)

OpenMesh – Mesh definition

```
#include <OpenMesh/Core/Mesh/Types/TriMesh_ArrayKernelT.hh>
typedef Openmesh::TriMesh_ArrayKernelT<>  Mesh;
```

name space

mesh type:

- triangle mesh
- array kernel
- default traits

OpenMesh – I/O

```
#include <OpenMesh/Core/IO/MeshIO.hh>
Mesh myMesh;
OpenMesh::IO::Options readOptions;
myMesh.request_vertex_colors();
...
If (! OpenMesh::IO::read_mesh(myMesh,
                              "/path/to/model",readOptions)) {
    std::cerr << "Error loading mesh." << std::endl;
    return 1;
  }
```

I/O options: Enable/got

- Vertex colors? Normals?…
- Texture coordinates?
- Face colors? Face normals?

OpenMesh – I/O Check result

```
if(!readOptions.check(OpenMesh::IO::Options::FaceNormal))
{
    myMesh->update_face_normals();
}
if(!readOptions.check(OpenMesh::IO::Options::VertexNormal))
{
    myMesh->update_vertex_normals();
}
```

OpenMesh - access

Access:

- All elements (faces, edges, halfedges and vertices) represented explicitly.
- Access through:
  - Handles (~ index)
  - Iterators (eg all vertices of the mesh)
  - Circular iterators (*circulators*): eg V:{F}

OpenMesh - access

Iterators and circulators:

- `Mesh::VertexIter`
- `Mesh::HalfEdgeIter`
- `Mesh::EdgeIter`
- `Mesh::FaceIter`

- `Mesh::VertexFaceIter`
- `Mesh::VertexVertexIter`
- `Mesh::VertexOHalfEdgeIter`
- `...`

OpenMesh - editing

```
using namespace MyMesh;

MyMesh mymesh;

// Add three vertices

VertexHandle v0 = mymesh.add_vertex(Point(0, 0, 0));

VertexHandle v1 = mymesh.add_vertex(Point(0, 1, 0));

VertexHandle v2 = mymesh.add_vertex(Point(3, 0, 2));

// Add the triangle

FaceHandle f = mymesh.add_face(v0, v1, v2);

// Delete a face

mymesh.delete_face(f);
```

OpenMesh - attributes

Adding attributes at runtime:

```
// Add a property to an edge
OpenMesh::EPropertyHandle<float> weight;
mymesh.add_property(weight);

// Setting a value
for ( MyMesh::EdgeIter e = mymesh.edges_begin();
                       e != mymesh.edges_end(); ++e )
    mymesh.property(weight, *e) = value;

// Getting a value: just use mymesh.property(weight, *e)
// as an r-value

// Remove the property
mymesh.remove property(weight);
```

OpenMesh – Fixed attributes

```
// General or triangle mesh? Which kernel?
#include <OpenMesh/Core/Mesh/TriMesh_ArrayKernelT.hh>
// Now choose types and attributes:
struct MyTraits : public OpenMesh::DefaultTraits
{
  // use double valued coordinates and normals
  typedef OpenMesh::Vec3d Point;
  typedef OpenMesh::Vec3d Normal;
  // use vertex normals and vertex colors
  VertexAttributes(OpenMesh::DefaultAttributer::Normal |
                   OpenMesh::DefaultAttributer::Color);
};

// finally construct the type of meshes we want:
typedef OpenMesh::TriMesh_ArrayKernelT<MyTraits> MyMesh;
...
```

References

- S. Campagna, L. Kobbelt, H.-P. Seidel, Directed Edges - A Scalable Representation For Triangle Meshes , ACM Journal of Graphics Tools 3 (4), 1998.

- Lutz Kettner, Using Generic Programming for Designing a Data Structure for Polyhedral Surfaces, in Proc. 14th Annual ACM Symp. on Computational Geometry, 1998.

- http://openmesh.org